

JABIR AHMAD JAHIN BHUIYAN

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🏛 EDUCATION

Islamic University of Technology (IUT)-OIC
Dhaka, Bangladesh

2022 – 2026

BSc. in Electrical and Electronic Engineering

CGPA: 3.94/4.00

Class Position: 6th out of 166

🔬 RESEARCH EXPERIENCE

Undergraduate Thesis

2025 – Present

- **Title:** DCA-DQG: Decentralized Context-Aware Dynamic Quantized Gossip for Communication-Efficient Traffic-State Estimation in Highway V2V Networks
- **Supervisor:** Prof. Dr. Mohammad Tawhid Kawser
- **Area/Field:** V2X Networks; Distributed Consensus; Dynamic Quantization
- **Description:** Developing communication-efficient distributed consensus algorithms for V2X networks using dynamic quantization and context-aware optimization for traffic-state estimation in highway V2V networks.

Independent Research Collaboration

2025 – Present

- **Title:** Learning-Driven Gossip Consensus for Communication-Constrained Distributed Networks
- **Supervisor:** Dr. Imtiaz Parvez, Utah Valley University (UVU), USA
- **Area/Field:** Distributed Networks; Gossip Consensus; Multi-Armed Bandits; Quantized Consensus
- **Description:** Investigating learning-driven gossip consensus algorithms for communication-constrained distributed networks. Co-authored three IEEE IETC 2026 papers on adaptive gossip consensus, quantized consensus, and smart-grid communication systems.

📖 PUBLICATIONS

1. **Jabir Ahmad Jahin Bhuiyan**, Md Mehedi Hasan, Syed Istiuk Raja, Kayd Craig, Mohammad T. Kawser, Imtiaz Parvez (2026). *An Adaptive Gossip Consensus Framework with UCB Multi-Armed Bandits for Limited-Bandwidth Networks*. 2026 Intermountain Engineering, Technology and Computing (IETC), IEEE.
<https://doi.org/10.1109/IETC69527.2026.11568654> **Status: Published**
2. Md Mehedi Hasan, **Jabir Ahmad Jahin Bhuiyan**, Syed Istiuk Raja, Jaxon Larsen, Mohammad Tawhid Kawser, Imtiaz Parvez (2026). *Learning-Driven Quantized Gossip Consensus via UCB Multi-Armed Bandits in Semi-Regular MGML Networks*. 2026 Intermountain Engineering, Technology and Computing (IETC), IEEE.
<https://doi.org/10.1109/IETC69527.2026.11568672> **Status: Published**
3. Syed Istiuk Raja, **Jabir Ahmad Jahin Bhuiyan**, Md Mehedi Hasan, Kayd Craig, Mohammad Tawhid Kawser, Imtiaz Parvez (2026). *A Learning-Driven Gossip Consensus for Smart Grid Applications*. 2026 Intermountain Engineering, Technology and Computing (IETC), IEEE.
<https://doi.org/10.1109/IETC69527.2026.11568655> **Status: Published**
Premium Technology Paper Award
4. MM Sadman Shafi, Tasnia Siddiqua Ahona, **Jabir Ahmad Jahin Bhuiyan** (2025). *Throughput Optimization via RL-Driven AMC with Dynamic Resource Scheduling in LTE*. 2025 IEEE Asia Pacific Conference on Wireless and Mobile (APWiMob), pp. 173–179.
<https://doi.org/10.1109/APWiMob67231.2025.11269118> **Status: Published**
5. **Jabir Ahmad Jahin Bhuiyan**, Mohammad Tawhid Kawser (2026). *Resource-Constrained Overlay–Underlay Mode Selection in LTE-V2X Networks Using Double Deep Q-Learning*. Wireless Personal Communications. **Status: Under Review**

6. Moontasir Rafique, M M Sadman Shafi, **Jabir Ahmad Jahin Bhuiyan**, Tasnia Siddiqua Ahona, Md Aadnan Farhad, Md. Samiur Rahman (2026). *Domain-Adversarial Adaptation for Robust RIS Beam Selection Under Propagation Shifts*. 2026 IEEE International Conference on Smart Connected Systems and Emerging Technologies (SCSETech). **Status: Under Review**
7. **Jabir Ahmad Jahin Bhuiyan**, Md Mehedi Hasan, Syed Istiuk Raja, Kayd Craig, Mohammad Tawhid Kawser (2026). *DCA-DQG: Decentralized Context-Aware Dynamic Quantized Gossip for Communication-Efficient Traffic-State Estimation in Highway V2V Networks*. Results in Engineering. **Status: Under Review**

SELECTED PROJECTS

5G System-Level Simulation and Performance Analysis

Documentation: [View Report](#)

- **Tools Used:** Vienna 5G Simulator, MATLAB
- **Objective:** Analyze user mobility, scheduling, feedback mechanisms, and link adaptation in a 5G system-level simulation.
- **Analysis:** Evaluated the impact of CQI feedback delay on throughput, effective SINR, feedback penalty, and BLER performance.

Wireless Channel Modeling and Analysis using NYUSIM

Documentation: [View Report](#)

- **Tools Used:** NYUSIM
- **Objective:** Model and analyze millimeter-wave Non-Line-of-Sight (NLOS) wireless channels under different antenna and propagation configurations.
- **Analysis:** Evaluated AOA, AOD, Power Delay Profile (PDP), received power, delay spread, and path loss.

Microwave MIMO Antenna Design and Optimization

Documentation: [View Report](#)

- **Objective:** Design and optimize a 2-element MIMO microstrip patch antenna operating at 6.9 GHz.
- **Design:** Implemented impedance matching, feed optimization, and Defected Ground Structure (DGS) to improve antenna isolation and performance.
- **Analysis:** Evaluated S-parameters, radiation characteristics, surface current distribution, and Envelope Correlation Coefficient (ECC).

SKILLS

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| ◦ Programming Languages <ul style="list-style-type: none">– Python, MATLAB, C, C++, Arduino IDE | ◦ Simulation & Engineering Tools <ul style="list-style-type: none">– MATLAB, Simulink, NYUSIM, Ansoft, Vienna 5G Simulator, CST Studio Suite, LINKPlanner |
| ◦ Machine Learning & AI <ul style="list-style-type: none">– PyTorch, Scikit-Learn, NumPy, Pandas, Reinforcement Learning, Deep Learning | ◦ Research & Development Tools <ul style="list-style-type: none">– LaTeX, Git, Academic Writing, Literature Review |

AWARDS & HONORS

- **Premium Technology Paper Award, i-ETC 2026** – for *Learning-Driven Quantized Gossip Consensus via UCB Multi-Armed Bandits in Semi-Regular MGML Networks*.
- **University Honors List** – CGPA above 3.75.
- **OIC Merit-Based Scholarship** – USD 14,000 merit scholarship.

REFERENCES

1. **Dr. Mohammad Tawhid Kawser**, Professor, Department of EEE, Islamic University of Technology (IUT), Bangladesh. kawser@iut-dhaka.edu
2. **Dr. Imtiaz Parvez**, Assistant Professor, Department of Computer Science, Utah Valley University (UVU), USA. imtiaz.parvez@uvu.edu